

Ishtiyak Ahmed

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EDUCATION

University of Toronto

*Honours Bachelor of Science in Geospatial Data Science;
Minors in Computer Science and Statistics*

Expected Apr. 2029

- Relevant Coursework: Software Design (Java), Programming (Python), Data Structures & Algorithms, Probability & Statistics, Theory of Computation

PROJECTS

Toronto Bike Share Operations | *Python, DuckDB, Parquet, GBFS*

Sep. 2026 – Present

- Built an initial 15-minute GBFS ingestion pipeline for Toronto Bike Share station inventory, validating API responses and implementing retry handling for recurring snapshots.
- Processed 15.3M+ historical bike-share trips with DuckDB and Parquet, standardizing heterogeneous schemas and aggregating station activity by station, weekday, and hour.
- Developing a historical-to-live operations analysis that will compare current station inventory against station-specific weekday/hour baselines to identify unusual availability patterns and support an interactive operational dashboard.

TTC Reliability | *Python, DuckDB, Parquet, Tableau, GIS*

Sep. 2026 – Present

- Engineered a Python/DuckDB/Parquet pipeline processing 175K+ TTC recorded delay records across four transit modes from Jan. 2025–Aug. 2026, with schema validation, duplicate quarantine, and SHA-256 change detection.
- Standardized recorded delay data by mode, route, weekday, hour, location, and delay category, conservatively matching 39K+ subway and 3.9K+ LRT records to official GTFS stops while preserving unmatched locations.
- Developed an interactive Tableau dashboard suite and GIS visualization to explore recorded delay activity; currently refining dashboard presentation and developing a case study to quantify recurring patterns, document methodology, and evaluate analytical limitations.

Othello Game Engine | *Java, JUnit 5, Git*

Oct. 2026

- Implemented core Othello functionality in Java, including board-state management, legal-move validation, piece flipping, turn management, and game progression.
- Developed AI gameplay and automated player matchups to evaluate different player strategies across human-versus-AI and AI-versus-AI games.
- Collaborated in a four-person Git workflow using parallel development, feature integration, code reviews, and JUnit-based testing.

Maze Solver and Visualizer | *Python, Pygame*

Aug. 2026

- Developed an interactive Pygame application that generates 2D/3D mazes from tree-based representations and uses preorder and level-order traversal to visualize start-to-finish paths.

File Hierarchy Visualizer | *Python, Pygame*

Mar. 2026

- Built a recursive treemap visualizer that maps file-system hierarchies and file sizes to interactive on-screen layouts with expandable, movable, resizable, and deletable nodes.

EXPERIENCE

Research and Operations Intern

Toronto, ON

E. Aferi

Feb. 2026 – Apr. 2026

- Conducted multi-source research and organized project findings, requirements, and reference materials into structured documentation for team decision-making.

TECHNICAL SKILLS

Languages: Python, Java, SQL

Data Technologies: DuckDB, Parquet, GBFS, Tableau, GIS

Data Methods: Data ingestion, schema validation, data cleaning, aggregation

Frameworks/Tools: JUnit 5, Pygame, Git, GitHub, Cursor, VS Code, Notion